

06_Advanced_SQL_Analysis

1: Monthly Revenue Trend Analysis

Business Question

How did monthly revenue change compared to the previous month?

SQL Technique Used

- Common Table Expression (CTE)
- Window Function
- LAG()

Key Findings

- March 2018 recorded the highest monthly revenue of 1,159,652.12.
- February 2018 generated revenue of 992,463.34.
- Revenue increased by 167,188.78 between February and March 2018.
- Revenue performance varied significantly across months, indicating changing customer purchasing behavior over time.

Business Insight

Monthly revenue trends reveal periods of accelerated growth and slowdown. Understanding these fluctuations helps identify seasonal demand patterns and supports more effective sales planning, inventory management, and marketing activities.

2: Revenue Growth Analysis

Business Question

How did revenue growth change from month to month?

SQL Technique Used

- Common Table Expression (CTE)
- Window Functions
- LAG()
- Percentage Growth Calculation

Key Findings

- February 2017 recorded revenue growth of 110.78% compared to January 2017.
- November 2017 achieved revenue growth of 53.25% compared to October 2017.
- December 2017 experienced the largest meaningful revenue decline of 26.49%.
- Revenue growth fluctuated throughout the analysis period, demonstrating periods of expansion and contraction.

Business Insight

Marketplace revenue growth was not linear. Monthly performance was influenced by changing customer demand patterns and seasonal purchasing behavior. Monitoring month-over-month growth provides early visibility into performance changes and helps support forecasting and strategic planning.

3: Seller Revenue Ranking

Business Question

Which sellers generate the highest marketplace revenue?

SQL Technique Used

- Common Table Expression (CTE)
- Window Functions
- RANK()

Key Findings

- The highest revenue-generating seller produced revenue of 507,166.91.
- The second-ranked seller generated 308,222.04.
- The third-ranked seller generated 301,245.27.
- A noticeable revenue gap exists between the top seller and the rest of the marketplace.

Business Insight

Seller performance is unevenly distributed across the marketplace. A small group of high-performing sellers contributes a disproportionately large share of revenue. Understanding the characteristics and practices of top-performing sellers can help improve overall marketplace performance.

4: Product Category Revenue Ranking

Business Question

Which product categories generate the highest marketplace revenue?

SQL Technique Used

- Common Table Expression (CTE)
- Window Functions
- DENSE_RANK()

Key Findings

- Bed, Bath & Table ranked first with revenue of 1,712,553.67.
- Beauty & Health ranked second with revenue of 1,657,373.12.
- Computer Accessories ranked third with revenue of 1,585,330.45.
- The top five categories generated substantial marketplace revenue and consistently outperformed other product segments.

Business Insight

Revenue generation is concentrated within a relatively small number of product categories. Home, lifestyle, and technology-related products represent the

strongest revenue drivers across the marketplace and play a critical role in overall business performance.

5: Cumulative Revenue Growth

Business Question

How did marketplace revenue accumulate over time?

SQL Technique Used

- Common Table Expression (CTE)
- Window Functions
- SUM() OVER()

Key Findings

- Total cumulative revenue reached 16,008,872.12.
- Revenue growth accelerated throughout 2017 and 2018.
- The marketplace exceeded 8 million cumulative revenue by January 2018.
- Cumulative revenue surpassed 12 million by May 2018.

Business Insight

Cumulative revenue analysis demonstrates consistent marketplace growth over time. The steady increase in total revenue indicates sustained customer demand and successful marketplace expansion throughout the observed period.

6: Top Revenue Month by Year

Business Question

Which month generated the highest revenue within each year?

SQL Technique Used

- Common Table Expression (CTE)
- Window Functions
- ROW_NUMBER()
- PARTITION BY

Key Findings

- October 2016 was the highest revenue month in 2016.
- November 2017 generated the highest revenue in 2017 with 1,194,882.80.
- April 2018 generated the highest revenue in 2018 with 1,160,785.48.
- November 2017 was the strongest revenue-performing month across the entire analysis period.

Business Insight

Identifying the best-performing month within each year helps reveal seasonal demand patterns and peak sales periods. These insights can support future planning for inventory, marketing campaigns, and resource allocation.

7: Revenue Variance Analysis

Business Question

Which months experienced the largest revenue increase and decrease compared to the previous month?

SQL Technique Used

- Common Table Expression (CTE)
- Window Functions
- LAG()
- Variance Analysis

Key Findings

- November 2017 recorded the largest revenue increase of 415,204.92 compared to the previous month.
- December 2017 recorded the largest revenue decline of 316,481.32 compared to the previous month.
- Revenue performance showed significant month-to-month fluctuations throughout the analysis period.

Business Insight

Revenue growth is not consistent across all periods. Certain months experience substantial increases or decreases in sales activity. Monitoring revenue variance helps identify peak demand periods, evaluate business performance, and improve forecasting accuracy.